

Multiple Choice (10 marks)

1. The Message confidentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Confidentiality
 - (c) Authentication
 - (d) Nonrepudiation
2. Which among the following is the least strong security encryption standard?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3
3. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web
4. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
 - (a) Insertion sort
 - (b) Quicksort
 - (c) Merge sort
 - (d) Selection sort
5. A/an _____ is a program that steals your logins & passwords for instant messaging applications.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) Trojan-Downloader
 - (d) Ransom Trojan

True / False (10 marks)

Answer True or False

6. True or False: Escaping queries is an effective method to prevent SQL injection by ensuring user input is treated as data rather than executable code.
7. True or False: A stack is a sequential segment of memory used for storing data such as character strings or arrays of integers.
8. True or False: Finding a shortest cycle in an undirected graph G is solvable in polynomial time.
9. True or False: A packet filter firewall functions at the application or transport layer, inspecting application data for security threats.
10. True or False: The four primary security principles related to messages are Confidentiality, Integrity, Non-repudiation, and Authentication.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the nested list elements which are present in another list.
12. Write a python function to find the maximum difference between any two elements in a given array.

End of Paper